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Roll No

EX-6005(2)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Determine the 8-point DFT of the following sequence using DIT FFT algorithm. $x(n) = \{1, 1, 2, 2, 3, 3, 4, 4\}$.
b) Compute the inverse DFT of $X(K) = \{1, 2, 3, 4, 5, 6\}$
2. a) Establish a relationship between DFT and Fourier series coefficients.
b) Explain the time shifting and frequency shifting property of DFT.
3. a) What is meant by impulse invariant method for designing IIR Filter?
b) Why impulse invariant method is not preferred in the design of IIR Filter other than low pass filter?
4. a) Compare Butterworth and Chebyshev filters.
b) Convert the analog filter with system function

$H(s) = \frac{s + 0.5}{(s + 0.5)^2 + 16}$ into a IIR filter using the bilinear transformation method. Let Resonant frequency.
 $\omega_r = \pi/2$

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5. a) Determine the normalized power and normalized energy.

i) $x(t) = e^{at} u(-t)$

ii) $x(t) = \cos(5\pi t) u(t)$

b) A DTS is described by the following expression

$y(n) = y^2(n-1) + x(n)$ with input $x(n) = 2\delta(n)$, check for the stability.

6. Find the frequency response and impulse response of LTI system characterized by difference equation

$$y(n) = \frac{3}{4} y(n-1) + \frac{1}{8} y(n-2) = 2x(n).$$

7. a) Prove that $\frac{d}{dt} x(t) \rightarrow j\omega x(\omega)$

b) Derive a relationship between input and output power spectral densities.

8. Explain the following (Any two)

a) Analog and digital filter

b) Bilinear transformation

c) Different window functions

d) Applications of Z Transform

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